

UNIT-2

LESSON 1: Introduction to Speech & Phonetics

Speech

Speech is the spoken form of language produced using human vocal organs.

Phonetics

Phonetics is the study of speech sounds and how they are:

Produced

Transmitted

Perceived

Types of Phonetics

Articulatory Phonetics

How sounds are produced (tongue, lips, vocal cords)

Acoustic Phonetics

Physical properties of sound

Frequency, pitch, amplitude

Auditory Phonetics

How humans hear and interpret sounds

Phone vs Phoneme

Phone: Actual spoken sound

Phoneme: Abstract sound unit that changes meaning

 Exam Focus:

Define phonetics + types (very common 5-mark question)

LESSON 2: Phonetic Transcription and IPA

Phonetic Transcription

Writing speech sounds using symbols instead of normal letters.

IPA (International Phonetic Alphabet)

Standard system for representing speech sounds

One symbol = one sound

Why IPA is important

Avoids spelling confusion

Helps in speech synthesis and recognition

Example

“cat” → /kæt/

 Exam Focus:

IPA definition + importance

LESSON 3: Speech Synthesis (Text-to-Speech)

Speech Synthesis

Process of converting text into spoken speech.

Main Steps

Text Analysis

Phoneme Generation

Prosody (stress, rhythm)

Waveform Generation

Types of Speech Synthesis

Concatenative

Uses stored speech units

Natural but large memory

Formant

Mathematical models

Less natural but flexible

Neural TTS

Uses deep learning

Very natural voice

Applications

Screen readers

Virtual assistants

Navigation systems

 Exam Focus:

Definition + types + block diagram

LESSON 4: Automatic Speech Recognition (ASR)

What is ASR?

Automatic Speech Recognition converts spoken words into text.

Basic ASR Components

Acoustic Model

Language Model

Pronunciation Dictionary

Decoder

ASR Working

Speech → Features → Models → Text output

Challenges

Noise

Accent

Continuous speech

 Exam Focus:

ASR block diagram + challenges

LESSON 5: Speech Recognition Models

Acoustic Model

Maps audio signals to phonemes

Language Model

Predicts word sequence probability

Pronunciation Model

Links words with phonemes

Decoding

Finds the most probable word sequence

 Exam Focus:

Explain models used in ASR

LESSON 6: Speech Recognition – Advanced Topics

1. Continuous Speech Recognition

Recognizes natural flowing speech

2. Speaker Dependent System

Trained for one speaker

3. Speaker Independent System

Works for many speakers

4. Large Vocabulary Speech Recognition

Thousands of words

Complex language models

5. Deep Learning in ASR

CNN, RNN, Transformers

High accuracy

 Exam Focus:

Short notes on advanced topics (any 3)

LESSON 7: Noise and Robust Speech Recognition

Problem of Noise

Background sound reduces accuracy

Solutions

Noise filtering

Feature normalization

Robust models

 Exam Focus:

Challenges + solutions

LESSON 8: Computational Phonology

Definition

Computational phonology uses computational techniques to study sound patterns in language.

Phonological Rules

Sound changes based on context

Example:

Cats → /s/

Dogs → /z/

Models Used

Finite State Transducers

Rule-based models

Constraint-based models

Applications

Speech recognition

Speech synthesis

Language learning systems